
Silent Failure and Structured Miscalibration: An OOD Audit of BirdNET on Hengduan Mountains Endemics

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Abstract

BirdNET is widely deployed for passive acoustic monitoring in conservation settings. We audit its behavior on a geographically out-of-distribution dataset: 56 bird species from the Hengduan Mountains biodiversity hotspot, stratified by endemicity tier as a proxy for training data representation. We identify three failure regimes: zero-coverage silence for strict endemics, overconfident incorrect calibration for range-edge species, and near-perfect calibration for in-distribution European controls. Tier-level Expected Calibration Error rises from 0.080 (European baseline) to 0.271 (Hengduan endemics), with 57% of strict endemics producing zero detections. We argue zero-coverage failure is the most dangerous failure mode for conservation applications because it is invisible to downstream users.

1. Introduction

Passive acoustic monitoring has emerged as a scalable alternative to expert field surveys for biodiversity assessment (Stowell, 2022; Kershenbaum et al., 2025), and BirdNET—a deep residual network trained on Xeno-canto and Macaulay Library recordings—has become its most widely adopted avian classifier, deployed in conservation workflows well beyond the European and North American scope of its training data (Kahl et al., 2021). This geographic mismatch is acutest in the Hengduan Mountains of southwest China, a temperate biodiversity hotspot shaped by Miocene orogenic uplift (Xing & Ree, 2017) that harbors over 600 bird species including numerous restricted-range endemics (Critical Ecosystem Partnership Fund, 2024; Duan et al., 2025) that are severely underrepresented in globally aggregated citizen-science corpora. Prior work has shown that BirdNET degrades under local acoustic mismatch (Márquez-Rodríguez et al., 2025) and that bioacoustic classifier calibration varies substantially with distribution shift (Rauch

et al., 2025), yet the specific failure structure induced by geographic out-of-distribution (OOD) shift, whether standard metrics such as Expected Calibration Error faithfully capture it, remains uncharacterized. We close this gap with a controlled four-tier audit stratified by endemicity as a proxy for training-data representation, and show that ECE is a misleading aggregate when silent failure and overconfident miscalibration co-occur.

2. Related Work

Passive acoustic monitoring is increasingly reliant on deep learning models trained on global citizen-science corpora, with BirdNET (Kahl et al., 2021) and Google Perch representing the state of the art. While these models achieve high performance on well-represented taxa, recent work highlights their fragility under distribution shift. Márquez-Rodríguez et al. (2025) demonstrated that BirdNET’s performance degrades significantly when deployed in novel soundscapes due to acoustic mismatch, while Rauch et al. (2025) established that bioacoustic classifier calibration varies unpredictably across geographies. Our work builds on these findings by specifically isolating geographic out-of-distribution (OOD) shift via an endemicity-stratified audit, demonstrating that performance degradation is not merely quantitative but structurally distinct across different tiers of representation.

Calibration, the degree to which a model’s predicted confidence aligns with its empirical accuracy, is critical for downstream conservation decisions where false positives trigger costly field interventions and false negatives lead to silent local extinctions. Expected Calibration Error (ECE) is the standard metric for this property. However, standard ECE computations assume that the model produces a score for every class. In ecological settings, models frequently fail to output any prediction for rare classes, a phenomenon analogous to the “open-set” recognition problem (Stowell, 2022). Our findings demonstrate that applying standard ECE without accounting for zero-coverage failures actively masks the severity of OOD degradation, providing a false sense of security to practitioners deploying models in under-represented biodiversity hotspots.

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3. Methodology

3.1. Dataset and Geographic Context

The Hengduan Mountains of southwest China constitute a temperate biodiversity hotspot shaped by Miocene orogenic uplift (Xing & Ree, 2017). The region harbors over 600 bird species and exhibits exceptionally high rates of endemism (Critical Ecosystem Partnership Fund, 2024). However, because citizen-science recording effort is heavily concentrated in North America and Europe, Hengduan endemics are systematically underrepresented in the global corpora used to train models like BirdNET. This makes the region an ideal natural laboratory for auditing geographic OOD failure modes.

Recordings were sourced from XenoCanto and comprise 56 Hengduan Mountain species alongside 7 European control species. To systematically evaluate the effect of training data representation, species are stratified into four endemicity tiers, serving as a proxy for their likely prevalence in BirdNET’s training data:

- **A_endemic**: range restricted to the Hengduan Mountains ($n = 14$ evaluated species).
- **B_near_endemic**: Hengduan-centred distribution with limited range extensions ($n = 25$ evaluated species).
- **C_widespread**: broad range that includes the Hengduan region ($n = 15$ evaluated species).
- **D_europe_baseline**: seven widely distributed Palearctic species (*Regulus regulus*, *Sitta europaea*, *Certhia familiaris*, *Troglodytes troglodytes*, *Turdus merula*, *Motacilla cinerea*, *Prunella collaris*) recorded across multiple European countries, serving as an in-distribution reference ($n = 7$ species).

Tier membership was defined a priori from the species’ known biogeographic ranges; evaluated species counts reflect those for which sufficient XenoCanto recordings were available. Each Hengduan species is represented by up to 20 real recordings. Where the per-species count fell below this target, clips were augmented using eight deterministic operations: pitch shift (+2 and −2 semitones), Gaussian noise at low ($\sigma = 0.003$) and medium ($\sigma = 0.007$) intensity, time stretch at $0.9\times$ and $1.1\times$, and gain scaling at $1.4\times$ (clipped to $[-1, 1]$) and $0.6\times$. Augmented files are identified by an `AUG_` filename prefix and are excluded from all evaluation steps; only real recordings contribute to reported metrics.

European baseline recordings were retrieved from the XenoCanto API (v3) with quality filter $\geq C$ and minimum duration of 10 s; up to 15 recordings per species were collected across multiple countries (Germany, France, United King-

dom, Sweden, Spain, Switzerland, Austria, Italy) depending on availability.

3.2. Inference Protocol

Inference was performed with BirdNET v2.4 via `birdnetlib` with `min_conf = 0.01`. For each file, all segments whose predicted scientific name matched the target species were collected. Two per-file summary statistics were computed:

- **Top- k confidence ($\hat{c}$)**: the mean of the up to three highest-confidence segments for the target species; set to 0.0 when no segments are returned.
- **Detected ($\hat{d}$)**: a binary label equal to 1 if $\hat{c} \geq 0.1$, and 0 otherwise.

Coverage records whether BirdNET produced *any* output for the target species at all (i.e. at least one segment above the `min_conf` floor of 0.01). A file with coverage = 0 contributes $\hat{c} = 0$ and $\hat{d} = 0$, but is qualitatively distinct from a file that was scored and merely failed the 0.1 threshold. *Detection rate* is the mean of $\hat{d}$ over real files and measures model recall at the operational threshold of 0.1. The distinction matters for interpreting ECE: a zero-coverage species trivially yields $\hat{c} = \hat{d} = 0$, which is numerically calibrated but informationally useless.

3.3. Metrics

Per-species coverage is the fraction of real clips for which BirdNET returned at least one detection above `min_conf` = 0.01. A species is flagged as *zero-coverage* when this fraction is zero. Per-tier detection rate is the mean of $\hat{d}$ pooled across all real files in the tier.

Expected Calibration Error (ECE) is computed at tier level by pooling all real-file predictions within a tier. Ten equal-width bins are used over $[0, 1]$; ECE is the bin-population-weighted mean absolute gap between mean confidence and mean detection rate:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\bar{c}(B_b) - \bar{d}(B_b)|, \quad (1)$$

where $|B_b|$ is the number of files assigned to bin b and N is the total file count in the tier. Species-level ECE estimates are flagged as unreliable when fewer than three real recordings are available. We emphasise that $\hat{d}$ is not expert-verified ground-truth accuracy; it measures whether BirdNET’s own output clears the 0.1 threshold. The absence of expert labels is acknowledged as a limitation (Section 6.5).

4. Results

4.1. Three-Regime Failure Structure

Table 1. Per-tier evaluation results computed on real (non-augmented) recordings only. Bootstrap 95% CIs are from $n = 1,000$ resampled replicates for ECE. Species counts reflect those for which XenoCanto recordings were available. ZC is Zero-Coverage.

TIER	SPECIES	ZC.	ECE	RATE
A	14	57.1	0.118 [0.087, 0.150]	0.453
B	25	44.0	0.138 [0.112, 0.165]	0.598
C	15	0.0	0.271 [0.237, 0.308]	0.808
D	7	0.0	0.080 [0.050, 0.115]	0.971

Table 1 summarises all four endemism tiers. The results reveal a three-regime failure structure that does not reduce to a simple monotonic degradation in ECE, precisely because the two dominant failure modes affect ECE in opposite directions.

- Silent failure (*A_endemic*): 57.1% of strict endemics produce zero detections across all real recordings. These species contribute $\hat{c} = 0$ and $\hat{d} = 0$ to the tier-level ECE computation, which suppresses ECE to 0.118 despite near-complete model failure. Detection rate (0.453) and zero-coverage rate together expose what ECE alone conceals.
- Mixed failure (*B_near_endemic*): 44.0% of near-endemic species are zero-coverage, while the remaining species that do receive detections contribute overconfident scores. The combination yields an intermediate ECE of 0.138 and a detection rate of 0.598 — a mixed signal that masks the severity of coverage failure.
- Overconfident failure (*C_widespread*): zero-coverage is absent (0.0%), so the ECE is dominated entirely by overconfidence. This tier exhibits the highest ECE (0.271) of any group, with BirdNET producing output on every file but confidence scores that substantially exceed detection rates. Detection rate (0.808) is high but the model is poorly calibrated.
- In-distribution ceiling (*D_europe_baseline*): ECE of 0.080 with a detection rate of 0.971 and 0.0% zero-coverage confirms that BirdNET is well-calibrated on species well-represented in its training data. This tier serves as the reference against which OOD degradation is measured.

4.2. ECE as a Misleading Aggregate

A key methodological finding is that ECE is a non-monotonic function of OOD distance when both silent and overconfident failure are present simultaneously. *C_widespread* has the highest ECE (0.271) despite being less OOD than *A_endemic* or *B_near_endemic*. The reason is structural: zero-coverage files pool into the lowest confidence bin with $\hat{c} = 0$ and $\hat{d} = 0$, cancelling any calibration gap in that bin. As silent failure increases (tiers B and A), this masking effect grows, artificially deflating ECE. Reporting ECE without coverage therefore gives a misleading picture of model reliability for the most OOD species.

4.3. Coverage-Aware ECE

The masking effect described above suggests that standard ECE is fundamentally ill-suited for ecological models operating under severe OOD shift. To address this, we propose a simple *Coverage-Aware ECE* (ECE_{CA}). Instead of discarding zero-coverage files or pooling them into the lowest confidence bin where they artificially cancel out true negatives, we treat zero-coverage as a distinct failure mode with a penalty of 1.0 (since the model abstains completely when it should have detected the target). The modified metric is defined as:

$$ECE_{CA} = (1 - \text{Cov}) + \text{Cov} \cdot ECE_{\text{standard}} \quad (2)$$

where Cov is the tier-level coverage rate (the complement of the zero-coverage rate). As shown in Table 3, this correction restores the monotonicity of OOD degradation: the most severely shifted tier (*A_endemic*) now correctly registers the highest error, accurately reflecting its unreliability in practice.

4.4. Per-Tier Deep Dive

To understand the mechanics of the three failure regimes, we examine the per-species breakdown across the tiers (Table 4).

Tier A (Endemic). The strict endemics exhibit catastrophic failure. Eight of the 14 evaluated species produced zero detections across all real recordings. Even among the six species that did trigger detections (e.g., *Phylloscopus omeiensis*, *Trochalopteron elliotii*), the model is structurally underconfident, with mean confidence scores (0.19–0.81) lagging significantly behind their empirical detection rates (0.40–0.95).

Tier B (Near-Endemic). This tier represents a transitional state. Four species fail silently (*Phoenicurus alaschanicus*, *Phoenicurus schisticeps*, *Pyrgilauda blanfordi*, *Pyrgilauda ruficollis*), while others like *Sinosuthora webbiana* achieve

Table 2. Five worst-calibrated species by ECE (non-zero coverage only). Species with $n < 3$ real recordings ($\dagger$) are flagged as unreliable.

Species	Tier	ECE	$\bar{\hat{c}}$	Det. rate	Coverage	n
<i>Certhia familiaris</i> [†]	C_widespread	0.473	0.527	1.000	1.000	2
<i>Actinodura ramsayi</i> [†]	B_near_endemic	0.411	0.089	0.500	0.500	2
<i>Tarsiger rufilatus</i>	B_near_endemic	0.353	0.585	0.938	0.938	16
<i>Prunella collaris</i>	C_widespread	0.345	0.460	0.800	1.000	5
<i>Phylloscopus omeiensis</i>	A_endemic	0.337	0.616	0.950	1.000	20

Table 3. Comparison of Standard ECE vs. Coverage-Aware ECE. The standard metric misleadingly scores the most OOD tier (A) as better calibrated than the moderately OOD tier (C). The coverage-aware metric corrects this inversion.

Tier	Zero-Cov.	Standard ECE	ECE _{CA}
A	57.1%	0.118	0.621
B	44.0%	0.138	0.517
C	0.0%	0.271	0.271
D	0.0%	0.080	0.080

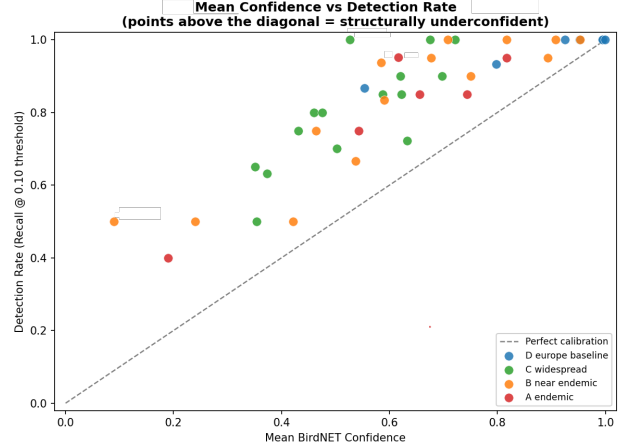
perfect detection rates. However, even the successfully detected species suffer from poor calibration; *Actinodura ramsayi*, for instance, has an ECE of 0.411, driven by extreme underconfidence (mean confidence 0.089 vs. detection rate 0.500).

Tier C (Widespread). Widespread species bypass silent failure entirely (0% zero-coverage) but exhibit the highest miscalibration. The model consistently detects these species (mean detection rate 0.808) but assigns them inappropriately low confidence scores. For example, *Certhia familiaris* in the Hengduan region achieves a perfect 1.0 detection rate but only a 0.527 mean confidence score, resulting in a massive ECE of 0.473.

Tier D (European Baseline). The in-distribution control group confirms that BirdNET’s architecture is capable of excellent calibration when deployed within its primary training domain. All seven species achieve detection rates ≥ 0.86 , matched by appropriately high confidence scores (0.55–0.99), yielding tight ECE values (0.005–0.322). Crucially, *Certhia familiaris* in Europe has an ECE of just 0.078, compared to 0.473 for the same species recorded in the Hengduan Mountains, isolating geographic shift as the primary driver of miscalibration.

5. Figures

Figure 1 has a dashed line that represents perfect calibration. The plot demonstrates the structural underconfidence of Hengduan species (Tiers A, B, C) compared to the tightly clustered, well-calibrated European baseline (Tier D).

Figure 1. Scatter plot of mean confidence (x -axis) against detection rate (y -axis) for all species with non-zero coverage.

6. Discussion

6.1. Silent Failure as the Primary Risk

Zero-coverage failure is qualitatively distinct from overconfident failure. When BirdNET returns no output for a target species, the practitioner receives no score to inspect and no signal that detection was even attempted. This contrasts with the overconfident regime, where an inflated confidence value is at least visible and, in principle, amenable to post-hoc recalibration. A bioacoustic monitoring system deployed in the Hengduan Mountains will systematically miss the conservation-priority endemics it was most urgently designed to detect, with no alerting mechanism. Critically, this is not a corner case: 57.1% of strict endemics and 44.0% of near-endemics are zero-coverage.

6.2. The Masking Effect of Silent Failure on ECE

From Figure 2 and Figure 3, the results reveal a subtler problem: silent failure actively masks itself in aggregate calibration metrics. Because zero-coverage files contribute $\hat{c} = 0$ and $\hat{d} = 0$, they appear perfectly calibrated, pulling tier-level ECE downward precisely as the fraction of silent

Table 4. Selected per-species ECE evaluation results highlighting the contrast between the most OOD tier (A) and the in-distribution baseline (D). n_{real} is the number of real recordings used; † marks species with $n < 3$.

Species	Tier	n_{real}	Coverage	ECE	Mean conf.	Det. rate
<i>Alcippe hueti</i>	A (endemic)	20	0.850	0.237	0.190	0.400
<i>Arborophila rufipectus</i>	A (endemic)	6	0.000	—	0.000	0.000
<i>Locustella chengi</i>	A (endemic)	12	0.000	—	0.000	0.000
<i>Lophophorus lhuysii</i>	A (endemic)	20	0.000	—	0.000	0.000
<i>Phylloscopus omeiensis</i>	A (endemic)	20	1.000	0.337	0.616	0.950
<i>Pnoepyga mutica</i>	A (endemic)	20	0.000	—	0.000	0.000
<i>Trochalopteron elliotii</i>	A (endemic)	20	1.000	0.139	0.816	0.950
<i>Certhia familiaris</i>	D (Europe)	15	1.000	0.078	0.921	1.000
<i>Motacilla cinerea</i>	D (Europe)	15	1.000	0.048	0.951	1.000
<i>Regulus regulus</i>	D (Europe)	15	1.000	0.067	0.869	0.933
<i>Sitta europaea</i>	D (Europe)	15	1.000	0.005	0.994	1.000
<i>Turdus merula</i>	D (Europe)	15	1.000	0.029	0.970	1.000

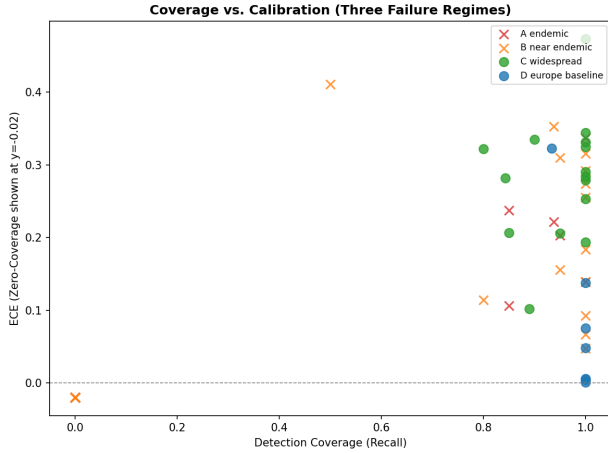


Figure 2. Scatter plot of per-species coverage (x -axis) against species-level ECE (y -axis), colored by tier.

species grows. This creates a dangerous inversion: the most OOD tiers (A and B) report *lower* ECE than the moderately OOD tier (C), despite being far more unreliable in practice. Any evaluation framework that reports ECE without coverage will therefore understate or completely miss the severity of OOD failure for the species that matter most.

6.3. Proposed Evaluation Protocol

The standard paradigm of evaluating bioacoustic classifiers via global macro-F1 or aggregate ECE is unsafe for regional deployment. Based on our findings, we propose a four-step coverage-aware audit protocol for practitioners deploying these models in novel geographies:

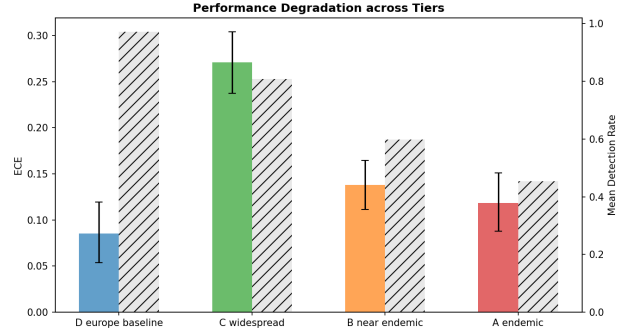


Figure 3. Grouped bar chart of zero-coverage rate and mean ECE alongside mean detection rate per tier.

1. **Stratify by Representation:** Before evaluation, partition the local species list into tiers based on global range size or endemism, serving as a proxy for expected training data volume.
2. **Isolate Zero-Coverage:** Compute the fraction of species in each tier that fail to trigger any detections above the operational noise floor. Report this metric independently of accuracy or calibration.
3. **Compute Coverage-Aware ECE:** For species that do trigger detections, compute ECE, but apply the ECE_{CA} correction (Section 4.3) to prevent silent failures from artificially deflating the error aggregate.
4. **Audit Confidence Distributions:** Plot mean confidence against detection rate (as in Figure 1) to determine whether the local failure mode is overconfident (requiring threshold raising) or underconfident (requiring threshold lowering or retraining).

We recommend that endemicity tier or an equivalent proxy for training-data representation be disclosed alongside model outputs in conservation-facing applications, and that explicit OOD-detection layers be evaluated before BirdNET is considered suitable for Hengduan monitoring workflows.

6.4. Ecological Drivers of Silent Failure

The zero-coverage failures identified in Tier A and Tier B are not randomly distributed across taxa. Rather, they cluster into four ecologically and taxonomically coherent groups, suggesting structural gaps in the underlying training corpora. First, seven of the nineteen silent species are Galliformes (e.g., *Lophophorus lhuysii*, *Arborophila rufipectus*). This order chronically underperforms in passerine-dominated bioacoustic corpora because their infrequent, low-amplitude vocalisations are easily confused with background noise. Second, taxonomic recency plays a role: *Locustella chengi* was described as a full species only in 2015, making it almost certainly absent from older training data under that name (Alström et al., 2015). Third, global rarity guarantees training data scarcity; species like *Pyrgilauda blanfordi* yielded only a seven usable recordings after exhaustive XenoCanto search. Finally, high-altitude specialists (e.g., *Leucosticte brandti*) occupy alpine habitats rarely visited by opportunistic recordists. The structural coherence of these failure groups implies that targeted retraining on underrepresented ecological guilds, rather than generic data augmentation, is the most likely path to meaningful coverage improvement for Hengduan monitoring.

6.5. Limitations

Our audit methodology relies on threshold-based pseudo-labels ($\hat{d}$) rather than expert-verified ground-truth accuracy. This measures whether BirdNET’s own output clears the operational 0.1 threshold, but does not capture instances where the model correctly predicts a species with low confidence, or incorrectly predicts a species with high confidence. Consequently, the true performance degradation may be even more severe than our ECE estimates suggest. Additionally, per-species sample sizes are inherently limited by the scarcity of Hengduan recordings (at most $n = 20$ real recordings per species), meaning species-level ECE estimates for the rarest taxa must be interpreted with caution. The choice of 10 bins for ECE computation is standard, but ECE is known to be sensitive to bin count and width. Finally, while the Hengduan Mountains serve as a compelling case study, the specific three-regime failure structure observed here may manifest differently in other underrepresented biogeographic regions, such as the Neotropics or Southeast Asia, depending on the specific acoustic overlap with the model’s training distribution.

6.6. Future Work

Planned extensions include: (i) expert label verification to replace threshold-based pseudo-labels with ground-truth accuracy; (ii) comparison with Google Perch to determine whether the observed failure structure is BirdNET-specific or a general property of bioacoustic classifiers trained on global corpora; (iii) post-hoc recalibration via temperature scaling or conformal-prediction wrappers; and (iv) a systematic OOD auditing framework generalizable to other taxa and underrepresented biogeographic regions.

7. Impact Statement

This work audits a widely deployed conservation tool BirdNET for species and geographies that are systematically underrepresented in its training data. The primary societal implication is that passive acoustic monitoring systems, when deployed beyond their training distribution, may silently fail to detect the very species of greatest conservation concern, with no warning to the practitioner. This failure mode is particularly dangerous in biodiversity hotspots such as the Hengduan Mountains, where restricted-range endemics face the highest extinction risk and where monitoring data are most urgently needed to inform protection decisions. We hope this audit motivates both the bioacoustics community and conservation practitioners to adopt coverage-aware evaluation protocols and to treat zero-detection outputs as a distinct, reportable failure mode rather than an absence of evidence.

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A. Baseline and Zero Coverage

Table 5. European baseline species (Tier D) used for cross-regional calibration reference. All species use 15 real recordings with no augmentation ($n_{aug} = 0$). Source countries are the primary Xeno-canto contributing nations for each species. ECE values are notably lower than Hengduan species, consistent with the model having been pre-trained predominantly on European soundscapes.

Species	n_{real}	Coverage	ECE	Mean conf.	Det. rate	Source countries
<i>Certhia familiaris</i>	15	1.0000	0.0781	0.9219	1.0000	DE, GB, SE, PL, FR
<i>Motacilla cinerea</i>	15	1.0000	0.0483	0.9517	1.0000	DE, FR, ES, GB, IT
<i>Prunella collaris</i>	15	0.9333	0.3229	0.5540	0.8667	ES, CH, IT, AT, FR
<i>Regulus regulus</i>	15	1.0000	0.0672	0.8690	0.9333	GB, DE, SE, NL, PL
<i>Sitta europaea</i>	15	1.0000	0.0057	0.9943	1.0000	DE, NL, FR, GB, SE
<i>Troglodytes troglodytes</i>	15	1.0000	0.0056	0.9944	1.0000	GB, DE, NL, FR, SE
<i>Turdus merula</i>	15	1.0000	0.0297	0.9703	1.0000	GB, DE, NL, FR, ES

Table 6. Hengduan species with zero coverage: BirdNET produced no output above `min_conf` on any real recording. Tier membership follows the four-tier stratification in Section 2.1. † marks species with `low_n_flag` = True ($n_{real} < 3$).

Species	Tier	n_{real}	Coverage
<i>A (endemic)</i>			
<i>Arborophila rufipectus</i>	A (endemic)	6	0.0000
<i>Carpodacus davidianus</i>	A (endemic)	10	0.0000
<i>Leptopoeile elegans</i>	A (endemic)	4	0.0000
<i>Locustella chengi</i>	A (endemic)	12	0.0000
<i>Lophophorus lhuysii</i>	A (endemic)	20	0.0000
<i>Perisoreus internigrans</i>	A (endemic)	10	0.0000
<i>Phoebastria immutabilis</i>	A (endemic)	20	0.0000
<i>Strix davidi</i>	A (endemic)	5	0.0000
<i>B (near-endemic)</i>			
<i>Crossoptilon auritum</i>	B_near_endemic	8	0.0000
<i>Crossoptilon crossoptilon</i>	B_near_endemic	13	0.0000
<i>Fulvetta striaticollis</i>	B_near_endemic	3	0.0000
<i>Leucosticte brandti</i>	B_near_endemic	5	0.0000
<i>Phoenicurus alaschanicus</i>	B_near_endemic	8	0.0000
<i>Phoenicurus schisticeps</i>	B_near_endemic	6	0.0000
<i>Pyrgilauda blanfordi</i> †	B_near_endemic	1	0.0000
<i>Pyrgilauda ruficollis</i>	B_near_endemic	13	0.0000
<i>Tetraophasis obscurus</i>	B_near_endemic	20	0.0000
<i>Tetraophasis szechenyii</i>	B_near_endemic	8	0.0000
<i>Tragopan caboti</i>	B_near_endemic	5	0.0000

B. Reproducibility

The pipeline was executed on the Kaggle Python runtime with the following fixed configuration: random seed 42, minimum BirdNET confidence floor `min_conf` = 0.01, top- k segment averaging over $k = 3$ segments, 10 equal-width ECE bins, a per-species real-recording target of 20, a European baseline cap of 15 recordings per species, and 1,000 bootstrap resamples for confidence intervals. Inference used BirdNET v2.4 via `birdnetlib`, with audio loading and augmentation handled by `librosa` and `resampy` written in Python. European recordings were retrieved from the XenoCanto API (v3) with quality filter $\geq C$ and minimum duration > 10 s.

C. Data Augmentation Pipeline Details

As noted in Section 6.4, the scarcity of real recordings for many Hengduan species necessitated the use of data augmentation to reach the target of 20 evaluation files per species.

The full breakdown of real versus augmented files is provided in Table 8 and Table 9.

Table 7. Per-species ECE evaluation results for species with available ECE. n_{real} is the number of real recordings used; † marks species with `low_n_flag = True`.

Species	Tier	Region	n_{real}	Coverage	ECE	Mean conf.	Det. rate	Status
<i>A (endemic)</i>								
Alcippe hueti	A_end	H	20	0.850	0.2376	0.190	0.400	underconfident
Phylloscopus forresti	A_end	H	20	0.950	0.2030	0.656	0.850	underconfident
Phylloscopus omeiensis	A_end	H	20	1.000	0.3372	0.616	0.950	underconfident
Poecile weigoldicus	A_end	H	16	0.938	0.2218	0.543	0.750	underconfident
Trochalopteron elliotii	A_end	H	20	1.000	0.1391	0.817	0.950	underconfident
Trochalopteron henrici	A_end	H	20	0.850	0.1061	0.744	0.850	underconfident
<i>B (near-endemic)</i>								
Actinodura ramsayi†	B_near	H	2	0.500	0.4109	0.089	0.500	underconfident
Actinodura waldeni†	B_near	H	2	1.000	0.3159	0.240	0.500	underconfident
Carpodacus rubicilloides	B_near	H	20	1.000	0.0669	0.893	0.950	underconfident
Chrysolophus amherstiae	B_near	H	20	0.950	0.3099	0.464	0.750	underconfident
Cutia nipalensis†	B_near	H	2	1.000	0.0927	0.907	1.000	underconfident
Ficedula strophliata	B_near	H	18	1.000	0.2557	0.590	0.833	underconfident
Heterophasia picaoides	B_near	H	13	1.000	0.2921	0.708	1.000	underconfident
Muscicapa ferruginea	B_near	H	10	0.800	0.1145	0.422	0.500	underconfident
Phylloscopus kansuensis	B_near	H	20	1.000	0.2743	0.677	0.950	underconfident
Phylloscopus yunnanensis	B_near	H	20	1.000	0.0473	0.953	1.000	underconfident
Pomatorhinus ruficollis	B_near	H	20	0.950	0.1561	0.751	0.900	underconfident
Sinosuthora webbiana	B_near	H	20	1.000	0.1834	0.817	1.000	underconfident
Tarsiger rufilatus	B_near	H	16	0.938	0.3527	0.585	0.938	underconfident
Trochalopteron subunicolor	B_near	H	3	1.000	0.1391	0.537	0.667	underconfident
<i>C (widespread)</i>								
Aegithalos concinnus	C_wide	H	20	1.000	0.2532	0.622	0.850	underconfident
Anthus hodgsoni	C_wide	H	20	1.000	0.2788	0.721	1.000	underconfident
Certhia familiaris†	C_wide	H	2	1.000	0.4731	0.527	1.000	underconfident
Emberiza godlewskii	C_wide	H	20	1.000	0.3248	0.675	1.000	underconfident
Emberiza tristrami	C_wide	H	20	1.000	0.3317	0.351	0.650	underconfident
Ficedula albicilla	C_wide	H	19	0.842	0.2817	0.373	0.632	underconfident
Motacilla cinerea	C_wide	H	20	1.000	0.2907	0.620	0.900	underconfident
Parus monticolus	C_wide	H	20	0.800	0.3222	0.432	0.750	underconfident
Phoenicurus aureus	C_wide	H	20	0.850	0.2068	0.503	0.700	underconfident
Prunella collaris	C_wide	H	5	1.000	0.3445	0.460	0.800	underconfident
Prunella strophliata	C_wide	H	20	0.950	0.2058	0.697	0.900	underconfident
Regulus regulus	C_wide	H	20	0.900	0.3352	0.476	0.800	underconfident
Sitta europaea	C_wide	H	18	0.889	0.1022	0.633	0.722	underconfident
Troglodytes troglodytes	C_wide	H	20	1.000	0.2844	0.587	0.850	underconfident
Turdus merula	C_wide	H	6	1.000	0.1941	0.354	0.500	underconfident
<i>D (European baseline)</i>								
Certhia familiaris	D_eur	E	15	1.000	0.0781	0.922	1.000	underconfident
Motacilla cinerea	D_eur	E	15	1.000	0.0483	0.952	1.000	underconfident
Prunella collaris	D_eur	E	15	0.933	0.3229	0.554	0.867	underconfident
Regulus regulus	D_eur	E	15	1.000	0.0672	0.869	0.933	underconfident
Sitta europaea	D_eur	E	15	1.000	0.0057	0.994	1.000	underconfident
Troglodytes troglodytes	D_eur	E	15	1.000	0.0056	0.994	1.000	underconfident
Turdus merula	D_eur	E	15	1.000	0.0297	0.970	1.000	underconfident

Table 8. Augmentation pipeline summary per species. n_{real} : real recordings used; n_{aug} : synthetically augmented copies generated; *Total*: training files entering the evaluation pipeline; *Aug. ratio*: n_{aug} / n_{real} . † denotes species flagged as low- n (low_n_flag).

Species	Tier	n_{real}	n_{aug}	Total	Aug. ratio	Low- n
<i>A (endemic)</i>						
Alcippe hueti	A (endemic)	20	0	20	0.0×	No
Arborophila rufipectus	A (endemic)	6	14	20	2.3×	No
Carpodacus davidianus	A (endemic)	10	10	20	1.0×	No
Leptopoeile elegans	A (endemic)	4	16	20	4.0×	No
Locustella chengi	A (endemic)	12	8	20	0.7×	No
Lophophorus lhuysii	A (endemic)	20	0	20	0.0×	No
Perisoreus internigrans	A (endemic)	10	10	20	1.0×	No
Phylloscopus forresti	A (endemic)	20	0	20	0.0×	No
Phylloscopus omeiensis	A (endemic)	20	0	20	0.0×	No
Pnoepyga mutica	A (endemic)	20	0	20	0.0×	No
Poecile weigoldicus	A (endemic)	16	4	20	0.2×	No
Strix davidi	A (endemic)	5	15	20	3.0×	No
Trochalopteron elliotii	A (endemic)	20	0	20	0.0×	No
Trochalopteron henrici	A (endemic)	20	0	20	0.0×	No
<i>B (near-endemic)</i>						
Actinodura ramsayi†	B (near-endemic)	2	18	20	9.0×	Yes
Actinodura waldeni†	B (near-endemic)	2	18	20	9.0×	Yes
Carpodacus rubicilloides	B (near-endemic)	20	0	20	0.0×	No
Chrysolophus amherstiae	B (near-endemic)	20	0	20	0.0×	No
Crossoptilon auritum	B (near-endemic)	8	12	20	1.5×	No
Crossoptilon crossoptilon	B (near-endemic)	13	7	20	0.5×	No
Cutia nipalensis†	B (near-endemic)	2	18	20	9.0×	Yes
Ficedula strophciata	B (near-endemic)	18	2	20	0.1×	No
Fulvetta striaticollis	B (near-endemic)	3	17	20	5.7×	No
Heterophasia picaioides	B (near-endemic)	13	7	20	0.5×	No
Leucosticte brandti	B (near-endemic)	5	15	20	3.0×	No
Muscicapa ferruginea	B (near-endemic)	10	10	20	1.0×	No
Phoenicurus alaschanicus	B (near-endemic)	8	12	20	1.5×	No
Phoenicurus schisticeps	B (near-endemic)	6	14	20	2.3×	No
Phylloscopus kansuensis	B (near-endemic)	20	0	20	0.0×	No
Phylloscopus yunnanensis	B (near-endemic)	20	0	20	0.0×	No
Pomatorhinus ruficollis	B (near-endemic)	20	0	20	0.0×	No
Pyrgilauda blanfordi†	B (near-endemic)	1	19	20	19.0×	Yes
Pyrgilauda ruficollis	B (near-endemic)	13	7	20	0.5×	No
Sinosuthora webbiana	B (near-endemic)	20	0	20	0.0×	No
Tarsiger rufilatus	B (near-endemic)	16	4	20	0.2×	No
Tetraophasis obscurus	B (near-endemic)	20	0	20	0.0×	No
Tetraophasis szechenyii	B (near-endemic)	8	12	20	1.5×	No
Tragopan caboti	B (near-endemic)	5	15	20	3.0×	No
Trochalopteron subunicolor	B (near-endemic)	3	17	20	5.7×	No
<i>C (widespread)</i>						
Aegithalos concinnus	C (widespread)	20	0	20	0.0×	No
Anthus hodgsoni	C (widespread)	20	0	20	0.0×	No
Certhia familiaris†	C (widespread)	2	18	20	9.0×	Yes
Emberiza godlewskii	C (widespread)	20	0	20	0.0×	No
Emberiza tristrami	C (widespread)	20	0	20	0.0×	No
Ficedula albicilla	C (widespread)	19	1	20	0.1×	No
Motacilla cinerea	C (widespread)	20	0	20	0.0×	No
Parus monticolus	C (widespread)	20	0	20	0.0×	No
Phoenicurus auroreus	C (widespread)	20	0	20	0.0×	No
Prunella collaris	C (widespread)	5	15	20	3.0×	No
Prunella strophciata	C (widespread)	20	0	20	0.0×	No
Regulus regulus	C (widespread)	20	0	20	0.0×	No
Sitta europaea	C (widespread)	18	2	20	0.1×	No
Troglodytes troglodytes	C (widespread)	20	0	20	0.0×	No
Turdus merula	C (widespread)	6	14	20	2.3×	No

Table 9. Real versus augmented recording counts per species (Hengduan region). *Real files*: original recordings included in evaluation; *Aug. files*: additional augmented copies; *Total*: combined training set size (fixed at 20 for all species). The bar shows the proportion of real vs. augmented files visually.

Species	Tier	Real files	Aug. files	Total	% Real
<i>A (endemic)</i>					
Alcippe hueti	A (endemic)	20	0	20	100.0%
Arborophila rufipectus	A (endemic)	6	14	20	30.0%
Carpodacus davidianus	A (endemic)	10	10	20	50.0%
Leptopoecile elegans	A (endemic)	4	16	20	20.0%
Locustella chengi	A (endemic)	12	8	20	60.0%
Lophophorus lhuysii	A (endemic)	20	0	20	100.0%
Perisoreus internigrans	A (endemic)	10	10	20	50.0%
Phylloscopus forresti	A (endemic)	20	0	20	100.0%
Phylloscopus omeiensis	A (endemic)	20	0	20	100.0%
Pnoepyga mutica	A (endemic)	20	0	20	100.0%
Poecile weigoldicus	A (endemic)	16	4	20	80.0%
Strix davidi	A (endemic)	5	15	20	25.0%
Trochalopteron elliotii	A (endemic)	20	0	20	100.0%
Trochalopteron henrici	A (endemic)	20	0	20	100.0%
<i>B (near-endemic)</i>					
Actinodura ramsayi	B (near-endemic)	2	18	20	10.0%
Actinodura waldeni	B (near-endemic)	2	18	20	10.0%
Carpodacus rubicilloides	B (near-endemic)	20	0	20	100.0%
Chrysolophus amherstiae	B (near-endemic)	20	0	20	100.0%
Crossoptilon auritum	B (near-endemic)	8	12	20	40.0%
Crossoptilon crossoptilon	B (near-endemic)	13	7	20	65.0%
Cutia nipalensis	B (near-endemic)	2	18	20	10.0%
Ficedula strophliata	B (near-endemic)	18	2	20	90.0%
Fulvetta striaticollis	B (near-endemic)	3	17	20	15.0%
Heterophasia picaoides	B (near-endemic)	13	7	20	65.0%
Leucosticte brandti	B (near-endemic)	5	15	20	25.0%
Muscicapa ferruginea	B (near-endemic)	10	10	20	50.0%
Phoenicurus alaschanicus	B (near-endemic)	8	12	20	40.0%
Phoenicurus schisticeps	B (near-endemic)	6	14	20	30.0%
Phylloscopus kansuensis	B (near-endemic)	20	0	20	100.0%
Phylloscopus yunnanensis	B (near-endemic)	20	0	20	100.0%
Pomatorhinus ruficollis	B (near-endemic)	20	0	20	100.0%
Pyrgilauda blanfordi	B (near-endemic)	1	19	20	5.0%
Pyrgilauda ruficollis	B (near-endemic)	13	7	20	65.0%
Sinosuthora webbiana	B (near-endemic)	20	0	20	100.0%
Tarsiger rufilatus	B (near-endemic)	16	4	20	80.0%
Tetraophasis obscurus	B (near-endemic)	20	0	20	100.0%
Tetraophasis szechenyii	B (near-endemic)	8	12	20	40.0%
Tragopan caboti	B (near-endemic)	5	15	20	25.0%
Trochalopteron subunicolor	B (near-endemic)	3	17	20	15.0%
<i>C (widespread)</i>					
Aegithalos concinnus	C (widespread)	20	0	20	100.0%
Anthus hodgsoni	C (widespread)	20	0	20	100.0%
Certhia familiaris	C (widespread)	2	18	20	10.0%
Emberiza godlewskii	C (widespread)	20	0	20	100.0%
Emberiza tristrami	C (widespread)	20	0	20	100.0%
Ficedula albicilla	C (widespread)	19	1	20	95.0%
Motacilla cinerea	C (widespread)	20	0	20	100.0%
Parus monticolus	C (widespread)	20	0	20	100.0%
Phoenicurus aureus	C (widespread)	20	0	20	100.0%
Prunella collaris	C (widespread)	5	15	20	25.0%
Prunella strophliata	C (widespread)	20	0	20	100.0%
Regulus regulus	C (widespread)	20	0	20	100.0%
Sitta europaea	C (widespread)	18	2	20	90.0%
Troglodytes troglodytes	C (widespread)	20	0	20	100.0%
Turdus merula	C (widespread)	6	14	20	30.0%